

SECURITY FIRMWARE PLAN

Title Page

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Introduction

Firmware is an essential component of modern digital devices. It provides low-level control for hardware and helps ensure that devices operate securely and efficiently. Keeping firmware updated is important because updates often contain security patches, performance improvements, and protection against newly discovered vulnerabilities. This report examines firmware security on selected devices connected to a home network and presents recommendations for maintaining a secure firmware environment.

Step 1: Devices with Firmware

The following devices connected to the network use firmware to perform their core functions:

1. MTN Router (MF935)

- Firmware controls network connectivity, Wi-Fi functionality, and device security settings.

2. TECNO Smartphone (Android 14)

- Firmware supports hardware operations, security functions, and operating system performance.

Step 2: Firmware Vulnerabilities and Prevention Measures

Vulnerability 1: Outdated Firmware

Description:

Outdated firmware may contain known security vulnerabilities that attackers can exploit.

Prevention Measures:

- Enable automatic firmware updates when available.
- Regularly check for firmware updates from the manufacturer.
- Install updates promptly.

Vulnerability 2: Default Credentials

Description:

Many devices are shipped with default usernames and passwords that can be easily guessed by attackers.

Prevention Measures:

- Change default credentials immediately after device setup.
- Use strong and unique passwords.
- Enable additional security features where available.

Vulnerability 3: Unverified Firmware Sources

Description:

Installing firmware from unofficial sources can introduce malware or unauthorized modifications.

Prevention Measures:

- Download firmware only from official manufacturer websites.
- Verify firmware authenticity before installation.
- Avoid third-party firmware unless specifically approved by the manufacturer.

Step 3: Firmware Assessment of Network Devices

Device 1: MTN Router (MF935)

Hardware Version:

MF935-HW1.0

Software Version:

BD_MTNNGMF935V1.0.0B10

Assessment:

The router is operating with the installed software version provided by the manufacturer. The administration interface does not display the exact firmware update date. Regular monitoring for manufacturer updates is recommended.

Device 2: TECNO Smartphone**Android Version:**

14

Android Security Update:

2026-05-01

Google Play System Update:

2026-03-01

Kernel Build Date:

January 28, 2026

Assessment:

The smartphone is running Android 14 with a recent security update. The device appears to be receiving security patches and system maintenance updates, which helps reduce exposure to known vulnerabilities.

Step 4: Personal Hardware and Firmware Security Plan

To maintain firmware security across all connected devices, the following actions will be implemented:

- Check for firmware updates monthly.
- Enable automatic updates whenever available.
- Replace default passwords with strong unique credentials.
- Download firmware only from trusted manufacturer sources.
- Monitor security advisories from device manufacturers.
- Remove unsupported or outdated devices from the network.
- Perform periodic security reviews of all connected devices.

Conclusion

Firmware security plays a critical role in protecting devices and networks from cyber threats. The assessment identified key firmware-related risks and demonstrated practical measures to reduce exposure to vulnerabilities. By maintaining regular updates, using secure credentials, and following manufacturer recommendations, users can significantly improve the security of their devices and network environment.

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